

# Brain-Computer Interface Session 1: Comprehensive Analysis

**Overview** This deck summarises signal-quality metrics, neural dynamics, multifractal complexity and baseline classification results from eight Session 1 EEG files (S1...S8\_Session\_1.mat). All figures were generated in the accompanying notebook. You can paste these Markdown slides into any converter (e.g. Pandoc, Marp) to create a PowerPoint.

## 1 Data Set & Processing Pipeline

### 1.1 Session Content

- 8 subjects, 1 session each ( 180 trials)
- 64-channel EEG, 1 kHz, motor-imagery tasks – Left / Right hand – Up / Down movement
- Trial timing: –2 s ITI → cue 0 s → feedback 2-4 s

### 1.2 Python Workflow

1. Load .mat → extract **BCI.data**, **TrialData**
2. **Quality control** – Bad-channel variance >5 z – Trial peak >100  $\mu$ V
3. **Pre-processing** – High-pass 1 Hz – Alpha band-pass 8-14 Hz
4. ERD, PSD, MFDFA, LDA

## 2 Signal-Quality Findings

| Subject | Bad Ch (n) | Artifact %   |
|---------|------------|--------------|
| S1      | 0          | 0.2%         |
| S2      | 0          | 0.7%         |
| S3      | <b>1</b>   | <b>97.6%</b> |
| S4      | 1          | 6.9%         |
| S5      | 0          | 8.7%         |
| S6      | 0          | 4.0%         |
| S7      | 0          | 0.2%         |
| S8      | 0          | 2.4%         |

### Take-aways

- Six subjects show excellent quality (<10% artifacts).
- S3 is dominated by artifacts; downstream metrics interpreted with caution.

## 3 Event-Related Desynchronisation (C3, 8-14 Hz)

| Condition   | Peak ERD C3 | Peak ERD C4 | Interpretation                     |
|-------------|-------------|-------------|------------------------------------|
| Right hand  | −30%        | −24%        | Strong contralateral suppression   |
| Left hand   | −28%        | −21%        | Mirror image of right-hand pattern |
| Up (both)   | −15%        | −14%        | Bilateral but weaker               |
| Down (rest) | −10%        | −8%         | Minimal modulation (expected)      |

- Baseline −1 s to 0 s; feedback window 2-4 s.
- Curves averaged after robust length-matching interpolation.

#### 4 Spectral Characteristics

- **PSD (C3, C4)** shows classic 1/f slope plus alpha peaks at 10 Hz.
- Band-power box-plots: Delta<Theta<Alpha<Beta<Gamma as expected.
- No abnormal line-noise or high-beta contamination.

#### 5 Multifractal Detrended Fluctuation Analysis

- Scales 10-1,000 samples;  $q = -5...5$ .
- Mean  $H(q)$  across subjects:  $0.30 \rightarrow 0.57$ .
- Non-linear  $H(q)$  curve confirms multifractality (monofractal would be flat).
- Greater complexity (higher  $H$  range) correlates with lower artifact rates.

#### 6 Baseline Classification (LDA, Mean + Std Features)

| Subject     | Trials (LR) | LDA Accuracy |
|-------------|-------------|--------------|
| S1          | 120         | 0.50         |
| S2          | 122         | 0.48         |
| S3          | 118         | 0.52         |
| S4          | 121         | 0.48         |
| S5          | 119         | 0.54         |
| S6          | 120         | 0.49         |
| S7          | 120         | 0.53         |
| S8          | 120         | 0.52         |
| <b>Mean</b> | —           | <b>0.509</b> |

#### Implications

- Simple mean/STD features hover at chance.
- ERD and multifractal features are necessary for practical decoding.

## 7 Code-Block Explanations (Key Snippets)

### 7.1 Butterworth Filters

```
_b, _a = butter(4, [8/(SRATE/2), 30/(SRATE/2)], btype='band')
_bhp, _ahp = butter(4, 1/(SRATE/2), btype='high')
```

- 4 -order = 24 dB/octave slope.
- Band-pass used for artifact peak detection; high-pass for ERD baseline stability.

### 7.2 Bad-Channel Detector

```
data = np.stack(truncated_trials, axis=1)          # (chan, trial, time)
var = data.var(axis=(1,2))
z = (var - var.mean())/var.std()
bad = np.where(z > 5)[0]
```

- Variance across *all* trials flags consistently noisy electrodes.
- Z-score threshold 5  $p < 10^{-5}$ .

### 7.3 Robust ERD Routine

```
avg = np.mean(interpolated_sigs, 0)
env = np.abs(hilbert(filtfilt(b_alpha, a_alpha, highpass(avg))))**2
erd = 100*(env - env_baseline)/env_baseline
```

- Interpolation equalises trial length.
- Hilbert envelope gives instantaneous band-power.

### 7.4 Manual MFDFA Fallback

```
for s in scales:
    rms = [(seg - seg.mean()).var()**0.5 for seg in segments]
    Fq[q] = (np.mean(rms**q))**(1/q) if q else np.exp(np.mean(np.log(rms)))
H = np.polyfit(np.log2(scales), np.log2(Fq), 1)[0]
```

- Implements profile detrending and fluctuation function when `mfdfa_fixed` is absent.

### 7.5 LDA Evaluation

```
scaler = StandardScaler()
X = scaler.fit_transform(features)
scores = cross_val_score(LDA(), X, y, cv=5)
```

- Standardisation ensures each feature contributes equally.
- 5-fold CV avoids optimistic bias, given small sample.

## 8 Recommendations

1. **Enhanced Features:** – ERD time-frequency maps – Riemannian covariance – Fractal bandwidth
2. **Artifact Mitigation:** Apply ICA to remove eye/muscle bursts (critical for S3).
3. **Classifier Upgrade:** Compare CSP + LDA, CNN-EEGNet, and transformer pipelines.
4. **Complexity Biomarkers:** Use  $H(q)$  range as covariate to predict BCI learning rate.

## 9 Conclusion

- Data quality is strong for 7/8 subjects; one requires curation.
- Alpha ERD confirms clear sensorimotor engagement.
- EEG signals exhibit multifractal structure—important for adaptive decoding.
- Baseline LDA reaches ~51%; advanced features are required for control-grade accuracy.